

Reboot Sequence Synchronization

Enforcing Strict Boot-Time Execution Order

Across `dcm-agent` and `reboot-manager`

Change: `reboot-sequence-sync`

Date: 2026-05-06 | **Status:** Proposal — Team Review

Agenda

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1. Problem Statement

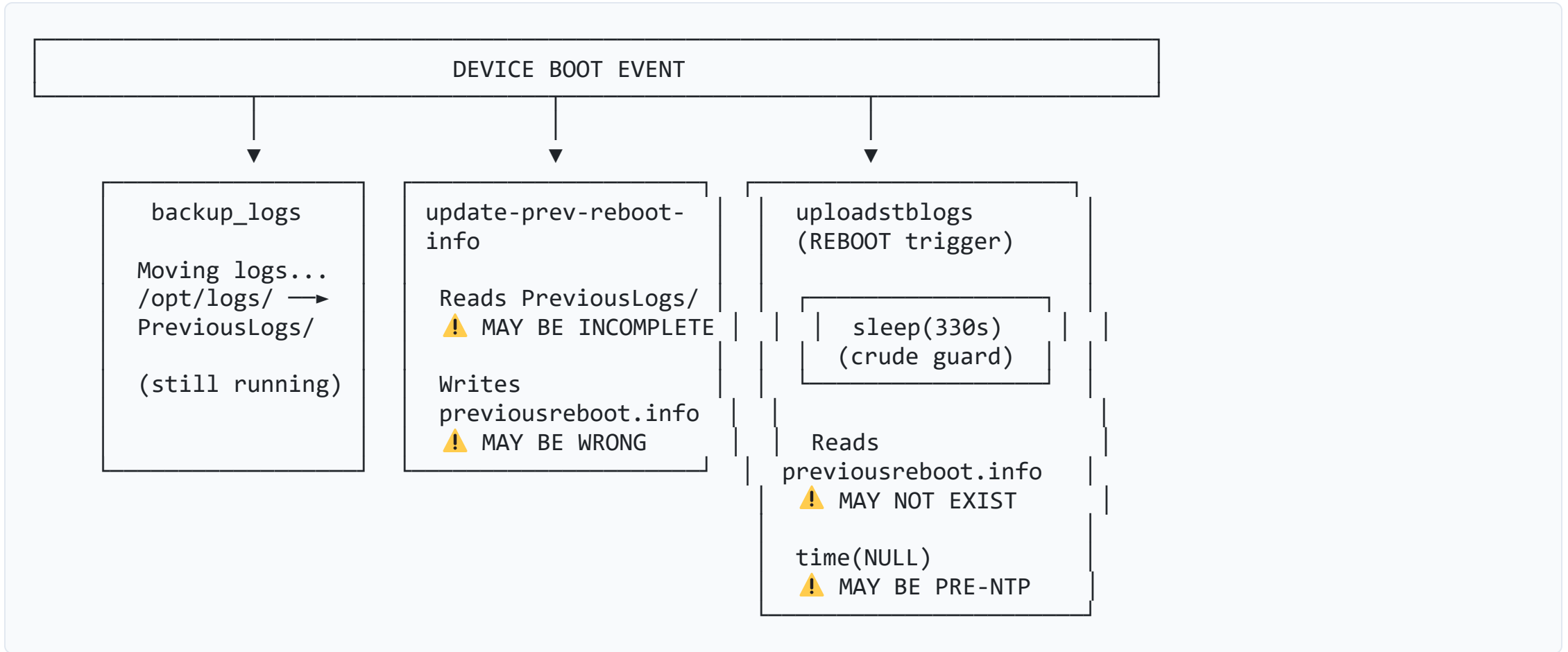
*As a device operator, I want log uploads after reboot to contain the **correct reboot reason** and **NTP-accurate archive timestamps**, so that support teams can reliably correlate uploaded logs with the actual reboot event.*

Three Boot-Time Subsystems — No Ordering Guarantee

Subsystem	Repository	Responsibility
backup_logs	dcm-agent	Moves live logs → PreviousLogs/
update-prev-reboot-info	reboot-manager	Derives reboot reason → writes previousreboot.info
uploadstblogs (REBOOT)	dcm-agent	Archives PreviousLogs/ → uploads to log server

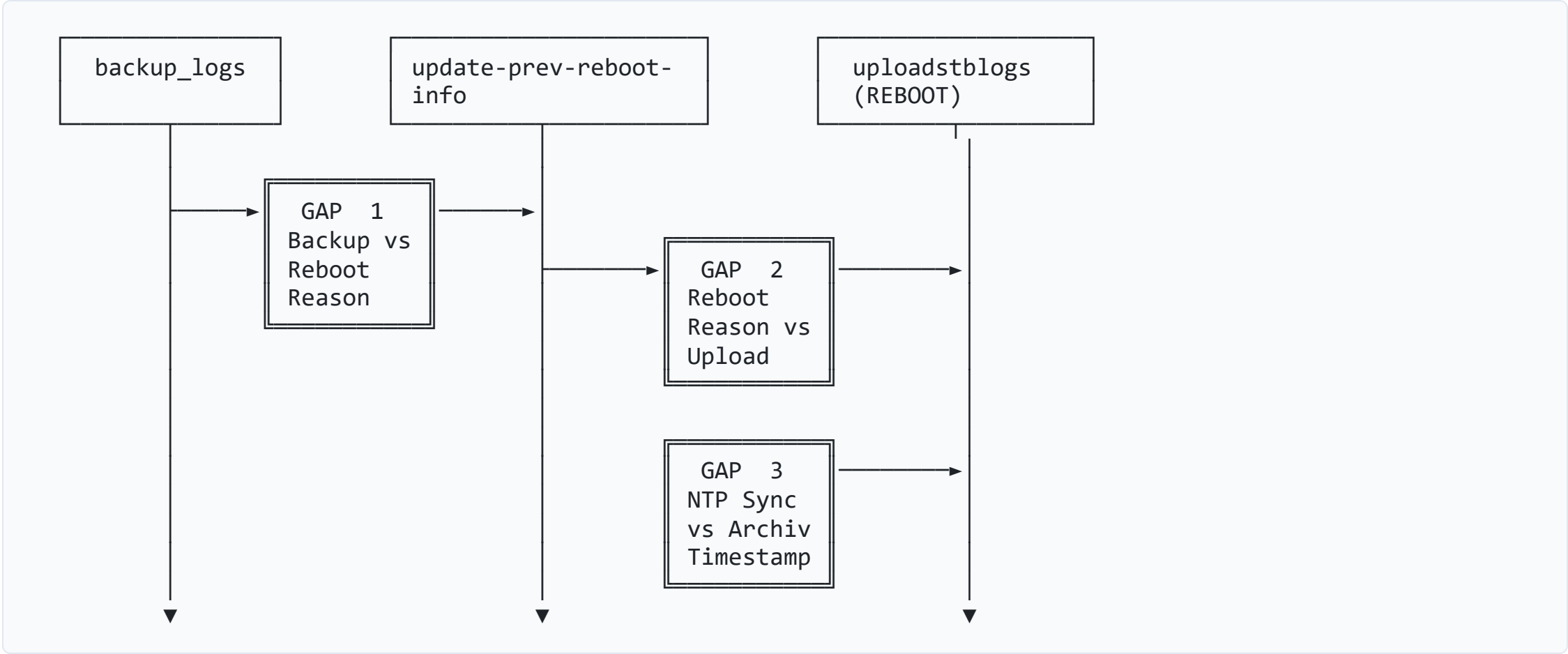
All three start independently at boot. **They race — and the race has real consequences.**

2. Current Boot Flow — The Race



No explicit ordering exists between any of these three processes.

3. Three Synchronization Gaps



Gap	Between	Risk
Gap 1	backup_logs → reboot-reason	Incomplete PreviousLogs/ → wrong reboot reason

4. The 330-Second Sleep Problem

The only existing "guard" is a crude sleep in `reboot_setup()` :

sleep(330) Analysis

Fast/Modern Device

backup_logs done in 2s
reboot-reason done in 5s
NTP sync done in 10s

uploadstblogs WAITS 330s anyway (5.5 min wasted!)

Slow/Loaded Device

backup_logs done in 120s
reboot-reason done in 200s
NTP sync done in 400s (slow network)

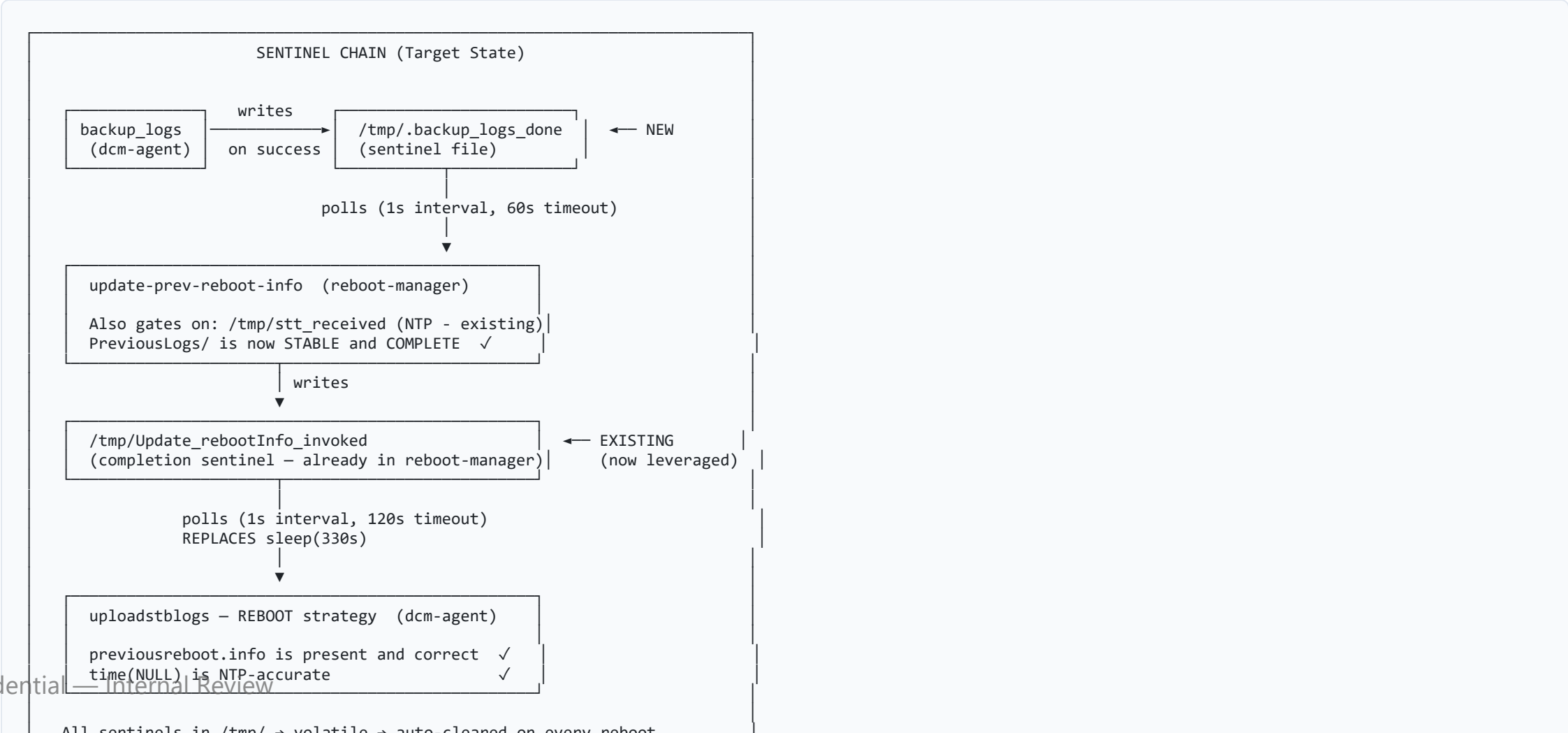
uploadstblogs STARTS at 330s – NTP NOT DONE!

Problem: sleep is NOT event-driven – fundamentally unreliable

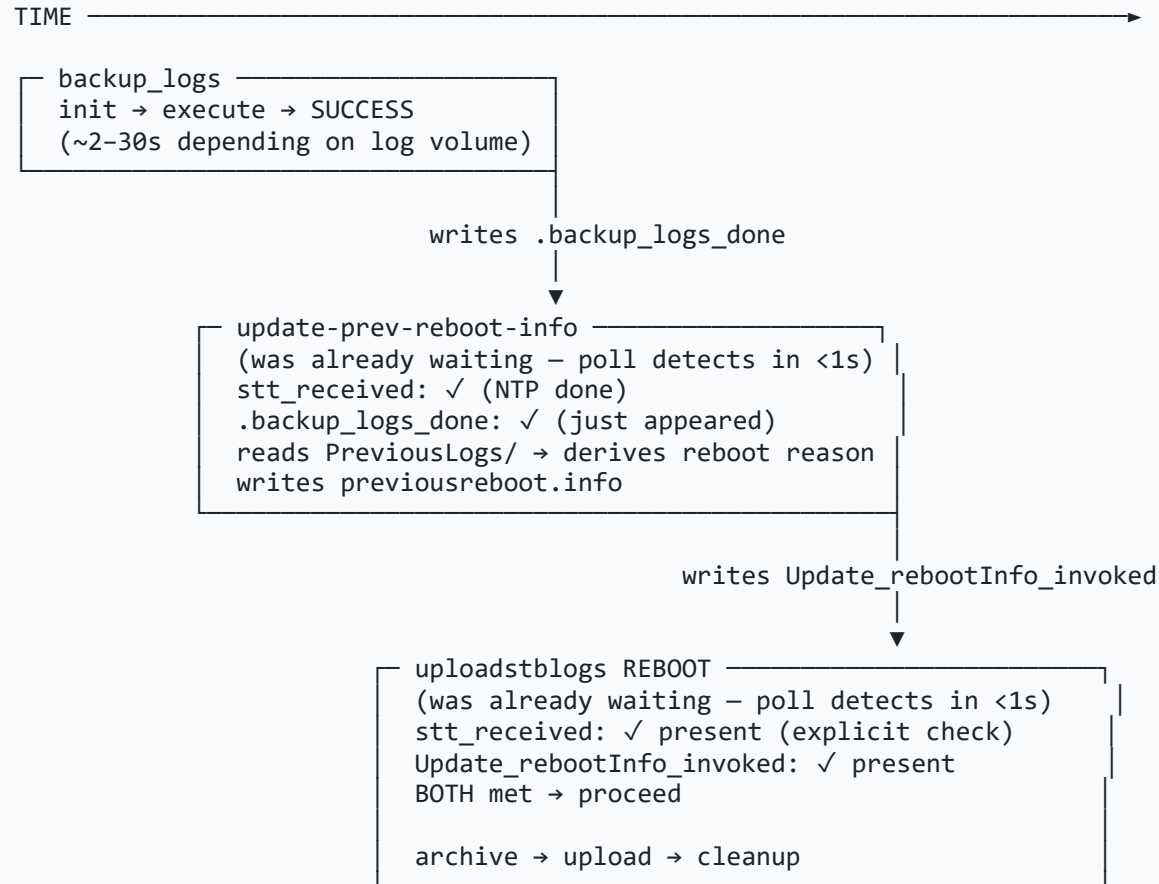
- Too long on fast devices → needlessly delays uploads
- Too short on slow devices → race conditions still happen
- No feedback mechanism → can't adapt to actual readiness

5. Proposed Solution — Sentinel Chain

Replace the sleep with **sentinel files** — lightweight, volatile, event-driven.



6. Happy Path — Timing Diagram



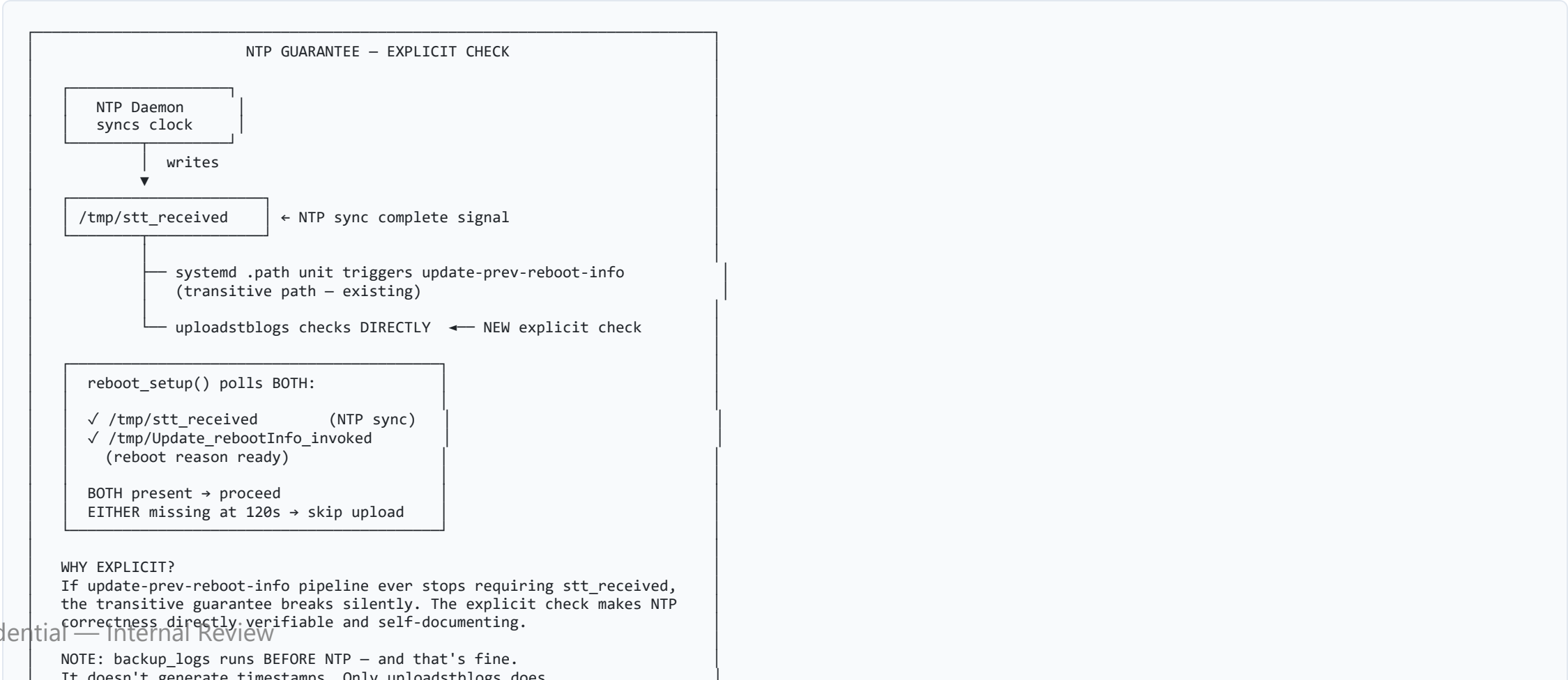
TOTAL WAIT: seconds (not 330s!)

Fast devices: upload starts within ~10s of boot

Slow devices: upload waits exactly as long as needed, no more

7. NTP Guarantee — Explicit Check

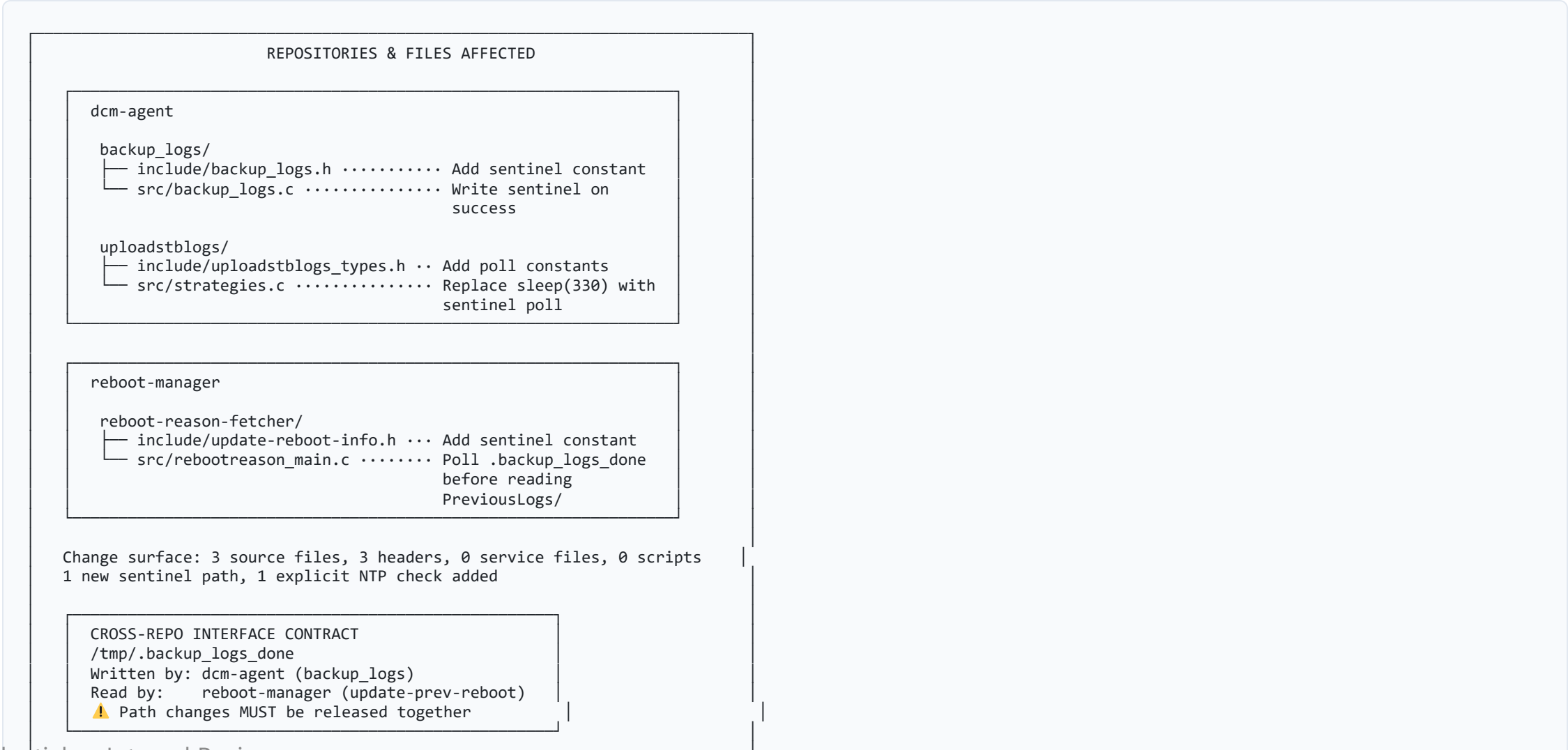
Key insight: `reboot_setup()` explicitly checks `/tmp/stt_received` alongside `/tmp/Update_rebootInfo_invoked` — belt-and-suspenders, not just transitive inference.



8. Key Design Decisions

#	Question	Decision	Rationale
1	Poll timeout for uploadstblogs?	120 s	Shorter than 330s; event-driven so fast devices proceed immediately
2	Poll interval?	1 s	Matches existing codebase pattern; low CPU overhead on embedded
3	Timeout action — uploadstblogs?	Skip upload	Safe degradation — no partial or mis-classified upload
4	Sentinel file location?	/tmp/	Volatile — auto-cleared on reboot; zero stale-sentinel risk
5	Timeout action — reboot-manager?	Exit with error	Prevents deriving wrong reboot reason from partial data
6	Specification scope?	Single spec	Both repos serve one boot pipeline; one spec = one truth
7	NTP for backup_logs?	Not required	Must run ASAP at boot; NTP only needed for archive naming (downstream)

9. Component Impact Map



10. Error Propagation & Timeout Behavior

FAILURE CASCADES CLEANLY

SCENARIO: backup_logs FAILS

backup_logs

- execute → FAILURE
- .backup_logs_done: X NOT written



update-prev-reboot-info

- polls .backup_logs_done for 60s
- TIMEOUT → logs error
- exits ERROR_GENERAL
- Update_rebootInfo_invoked: X NOT written



uploadstblogs

- polls Update_rebootInfo_invoked for 120s
- TIMEOUT → logs error
- returns -1 → strategy handler SKIPS upload
- NO partial upload, NO wrong data sent to server



Device continues boot normally
Next reboot: /tmp/ cleared → fresh start

KEY PROPERTY: Failure cascades are SAFE – no corrupt data, no partial uploads. The device simply skips the upload for this boot cycle.

11. Polling Approach — Why stat() ?

Criterion	inotify	stat() polling
POSIX portability	Linux-only	All target platforms
Embedded resource cost	Higher (kernel watch descriptors)	Minimal
Already in codebase?	No	Yes (dcm_utils.c pattern)
Code complexity	Higher (event loop, fd mgmt)	Simple loop
Timing accuracy	Event-exact	±1 s (acceptable)
Signal interruption risk	fd-based — complex	Benign (re-checks next iteration)

Decision: stat() polling — consistent with existing codebase, portable, minimal.

12. Testing Strategy

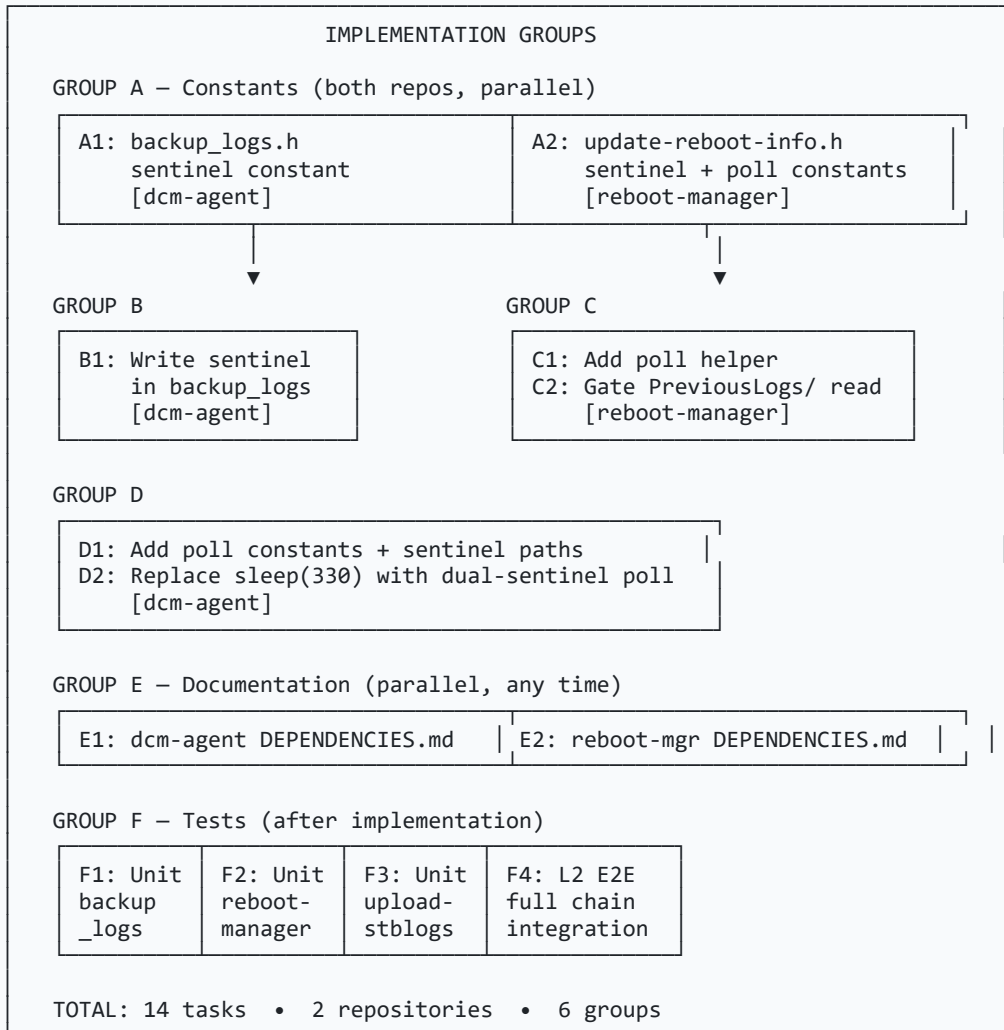
Unit Tests (per module)

ID	Verifies	Module	Type
TEST-SYNC-001	Sentinel written on success, absent on failure	backup_logs	GTest
TEST-SYNC-002	Proceeds when sentinel present	reboot-manager	GTest
TEST-SYNC-003	Exits error after timeout (no sentinel)	reboot-manager	GTest
TEST-SYNC-004	<code>reboot_setup()</code> proceeds when both sentinels present	uploadstblogs	GTest
TEST-SYNC-005	<code>reboot_setup()</code> skips upload if either sentinel missing at timeout	uploadstblogs	GTest
TEST-SYNC-006	Archive timestamp is post-NTP (mocked <code>time()</code>)	uploadstblogs	GTest

L2 Integration (Docker CI)

Step	Verification
Run backup_logs	<code>.backup_logs_done</code> created

13. Task Breakdown



14. What We Are NOT Changing

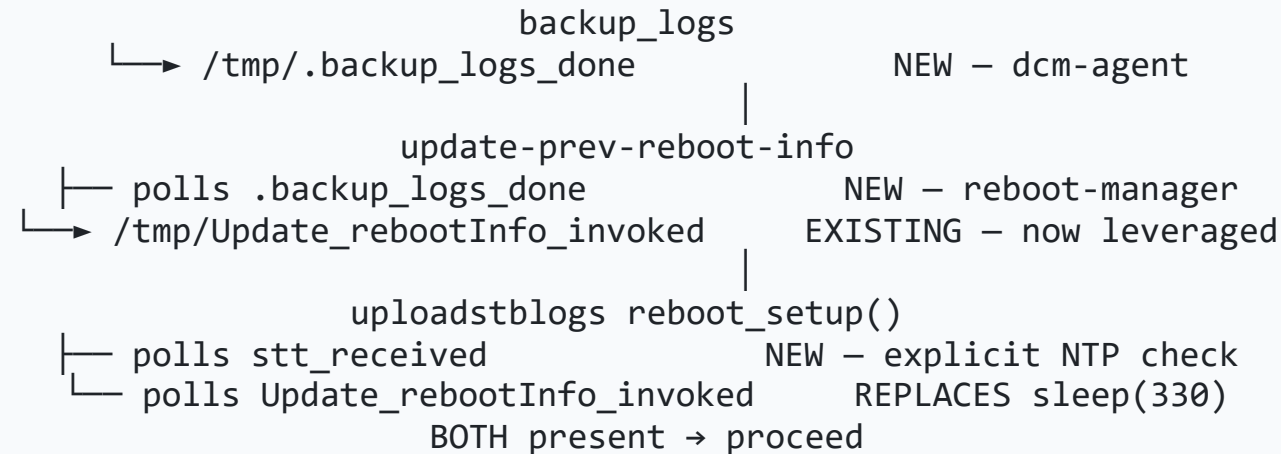
Area	In scope?	Reason
TRIGGER_DCM upload path	No	Only REBOOT trigger has the race
TRIGGER_ONDEMAND path	No	User-triggered; not a boot-sequence concern
Scheduled log upload	No	Different code path entirely
NTP subsystem	No	We consume its signal, don't change it
Systemd service/path units	No	Existing .service / .path files unchanged
backup_logs copy logic	No	Only sentinel write added at end
Shell scripts (legacy)	No	C binary paths only (future task for script parity)

15. Before vs. After

Dimension	Before	After
Ordering	None — three concurrent processes	Strict chain via sentinel files
Guard mechanism	<code>sleep(330s)</code> — fixed, blind	Event-driven poll — reacts to actual completion
Reboot reason	May be wrong (partial PreviousLogs/)	Always correct (backup confirmed done first)
Archive timestamp	May be pre-NTP	Guaranteed NTP-accurate (explicit <code>/tmp/stt_received</code> check)
Fast device boot	Wastes ~5 min sleeping	Upload starts within seconds of readiness
Slow device boot	May still race (330s not enough)	Waits exactly as long as needed
Failure mode	Undefined — may upload corrupt data	Clean: timeout → skip → log error → continue boot
backup_logs NTP dep	None	None (unchanged — NTP-free by design)

Questions & Discussion

SENTINEL CHAIN RECAP



Artifacts: `openspec/changes/reboot-sequence-sync/`

`proposal.md` — What and why | `spec.md` — Requirements (REQ-SYNC-001 –

007)